

# Bluestock Mutual Fund Analytics Capstone

BLUESTOCK MUTUAL FUND ANALYTICS CAPSTONE

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**Project Report**

**Submitted By**

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**Internship Program**

Bluestock Fintech

Mutual Fund Analytics Capstone Project

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**Technologies Used**

Python | SQL | SQLite | Power BI | Pandas | Jupyter Notebook

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**Submission Date**

June 2026

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**BLUESTOCK FINTECH**

2026

## Executive Summary

The Bluestock Mutual Fund Analytics Capstone Project was developed to create a comprehensive data analytics platform for analyzing mutual fund performance, investor behavior, portfolio composition, and market trends. The project integrates data engineering, database management, statistical analysis, advanced risk analytics, and business intelligence reporting to provide meaningful insights for investment decision-making.

The project utilized multiple datasets including mutual fund NAV history, fund master information, investor transaction records, portfolio holdings, and benchmark index data. The data was extracted, cleaned, transformed, and stored in a SQLite database using Python and SQL-based ETL processes. A structured database schema was designed to support efficient analytical queries and dashboard reporting.

Exploratory Data Analysis (EDA) was performed to understand trends in Assets Under Management (AUM), investor participation, fund categories, and transaction behavior. Key performance metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, Volatility, CAGR, and Maximum Drawdown were computed to evaluate mutual fund performance on a risk-adjusted basis.

Advanced analytics techniques were implemented to further enhance decision-making capabilities. Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR) were calculated to assess downside risk. Rolling 90-Day Sharpe Ratio analysis was conducted to evaluate the consistency of risk-adjusted returns over time. Investor Cohort Analysis and SIP Continuity Analysis were used to study investor retention and identify potentially inactive investors. Sector concentration risk was evaluated using the Herfindahl-Hirschman Index (HHI), while a risk-based fund recommendation system was developed to suggest suitable mutual funds based on investor risk profiles.

To enable interactive exploration of insights, a four-page Power BI dashboard was developed. The dashboard includes Market Overview, Fund Performance & Risk, Investor Analytics, and SIP & Market Trends sections. Interactive filters, KPI cards, charts, and scorecards were incorporated to support dynamic analysis and reporting.

The project successfully demonstrates the application of Python, SQL, SQLite, and Power BI in solving real-world financial analytics problems. The final solution provides a scalable framework for evaluating mutual fund performance, understanding investor behavior, assessing portfolio risk, and supporting data-driven investment decisions.

# Project Objectives

The primary objective of this project was to develop a comprehensive Mutual Fund Analytics Platform capable of transforming raw financial data into meaningful business insights. The project aimed to integrate data engineering, analytics, and business intelligence techniques to support informed investment decision-making.

The specific objectives of the project were:

1. To collect, clean, and integrate mutual fund data from multiple sources including NAV history, investor transactions, portfolio holdings, and benchmark indices.
2. To design and implement a structured SQLite database for efficient storage, retrieval, and analysis of mutual fund data.
3. To perform Exploratory Data Analysis (EDA) in order to identify trends, patterns, and relationships within mutual fund and investor datasets.
4. To evaluate mutual fund performance using key financial metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, Volatility, CAGR, and Maximum Drawdown.
5. To conduct advanced risk analysis using Value at Risk (VaR), Conditional Value at Risk (CVaR), and Rolling Sharpe Ratio techniques.
6. To analyze investor behavior through Cohort Analysis and SIP Continuity Analysis in order to understand investment patterns and retention trends.
7. To assess portfolio concentration and diversification using the Herfindahl-Hirschman Index (HHI).
8. To develop a risk-based mutual fund recommendation system capable of suggesting suitable funds for Low, Moderate, and High-risk investors.
9. To create an interactive Power BI dashboard that provides a visual representation of market trends, fund performance, investor analytics, and SIP inflows.
10. To generate actionable insights that can assist investors, fund managers, and financial analysts in making data-driven investment decisions.

By achieving these objectives, the project demonstrates the practical application of Python, SQL, SQLite, and Power BI in solving real-world financial analytics challenges.

# Data Sources

## 1. Fund Master Dataset

### Key Attributes

- AMFI Code
- Scheme Name
- Fund House
- Category
- Plan Type
- Risk Grade
- Morningstar Rating

## 2. NAV History Dataset

The Net Asset Value (NAV) dataset contains historical daily NAV values for mutual fund schemes. It forms the foundation for performance and risk analysis.

### Key Attributes

- AMFI Code
- Date
- NAV
- VaR and CVaR
- Rolling Sharpe Ratio

The NAV dataset is one of the most critical data sources in the project as it enables the evaluation of fund performance over time.

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## 3. Investor Transactions Dataset

The Investor Transactions dataset contains information regarding investments made by investors in different mutual fund schemes.

### Key Attributes

- Investor ID
- Transaction Date
- Amount Invested
- State
- City Tier
- Age Group
- Transaction Type
- State-wise investment distribution

#### **4. Portfolio Holdings Dataset**

##### **Key Attributes**

- AMFI Code
- Stock Symbol
- Stock Name
- Sector
- Weight Percentage
- Market Value
- Portfolio Date
- NIFTY500
- BSE Smallcap
- CRISIL Liquid
- CRISIL Gilt
- Support Alpha and Beta calculations

##### **Summary**

Together, these datasets create a comprehensive analytical framework that supports mutual fund performance evaluation, investor analytics, portfolio risk assessment, and business intelligence reporting. The integration of these datasets enables the development of a robust end-to-end Mutual Fund Analytics platform.

# ETL Pipeline

## ETL Pipeline

The ETL (Extract, Transform, Load) pipeline was designed to collect, clean, integrate, and store mutual fund data from multiple sources for further analysis and dashboard reporting.

## ETL Workflow

Raw Data Sources

(Fund Master, NAV, Transactions,

Portfolio Holdings, Benchmark Indices)

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Extract

|



Transform

(Data Cleaning, Validation,

Missing Value Handling,

Data Standardization)

|



Load

(SQLite Database)

|



Analytics & Reporting Layer

(Python Analysis + Power BI Dashboard)

## Extract

The extraction stage involved importing raw datasets from CSV files containing mutual fund information, NAV history, investor transactions, portfolio holdings, and benchmark index data. Python libraries such as Pandas were used to read and process the source files.

## **Transform**

During the transformation stage, the datasets were cleaned and standardized to ensure consistency and accuracy. This process included handling missing values, removing duplicate records, converting date formats, validating data types, creating calculated fields, and preparing the data for analysis.

Key transformations performed:

- Missing value treatment
- Duplicate removal
- Date standardization
- Column normalization
- Daily return calculation
- Risk metric preparation
- Dataset integration

## **Load**

The cleaned datasets were loaded into a SQLite database designed using a star-schema approach. Fact and dimension tables were created to support efficient analytical queries and reporting.

The final processed data was then utilized for:

- Exploratory Data Analysis (EDA)
- Performance Analytics
- Advanced Risk Analytics
- Investor Behavior Analysis
- Power BI Dashboard Development

The ETL pipeline ensured a reliable and structured data foundation for all subsequent analytics and business intelligence activities performed in the project.

# Database Design

## Database Design

The project uses a SQLite database designed using a star-schema approach to efficiently store and analyze mutual fund data. The database consists of one dimension table and multiple fact tables that support performance analytics, investor analysis, and dashboard reporting.

## Database Schema

### Tables Description

#### Dimension Table

##### **dim\_fund**

Stores master information about mutual fund schemes including scheme name, fund house, category, plan type, risk grade, and ratings.

#### Fact Tables

##### **fact\_nav**

Contains historical Net Asset Value (NAV) records used for return calculations, volatility analysis, and performance measurement.

##### **fact\_aum**

Stores Assets Under Management (AUM) information used for industry trend and fund size analysis.

##### **fact\_performance**

Contains calculated performance metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, Volatility, CAGR, and Maximum Drawdown.

##### **fact\_transactions**

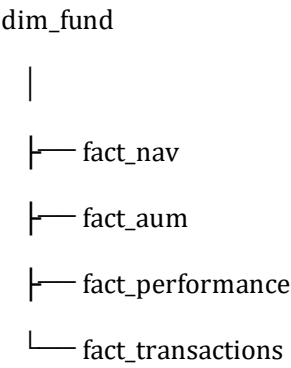
Stores investor transaction details including investment amount, transaction date, state, city tier, and investor demographics.

### Table Relationships



The dim\_fund table acts as the central dimension table and is linked to all fact tables through the **AMFI Code**. This relationship enables integrated analysis across NAV history, performance metrics, AUM trends, and investor transactions.

**Relationship Flow**



This structure improves query performance, reduces data redundancy, and supports efficient reporting in Power BI dashboards and analytical notebooks.

# Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure, quality, and characteristics of the mutual fund datasets before conducting advanced analytics. The analysis focused on identifying trends in fund performance, investor participation, and asset distribution.

## Visualizations

Chart 1: Monthly sip inflow

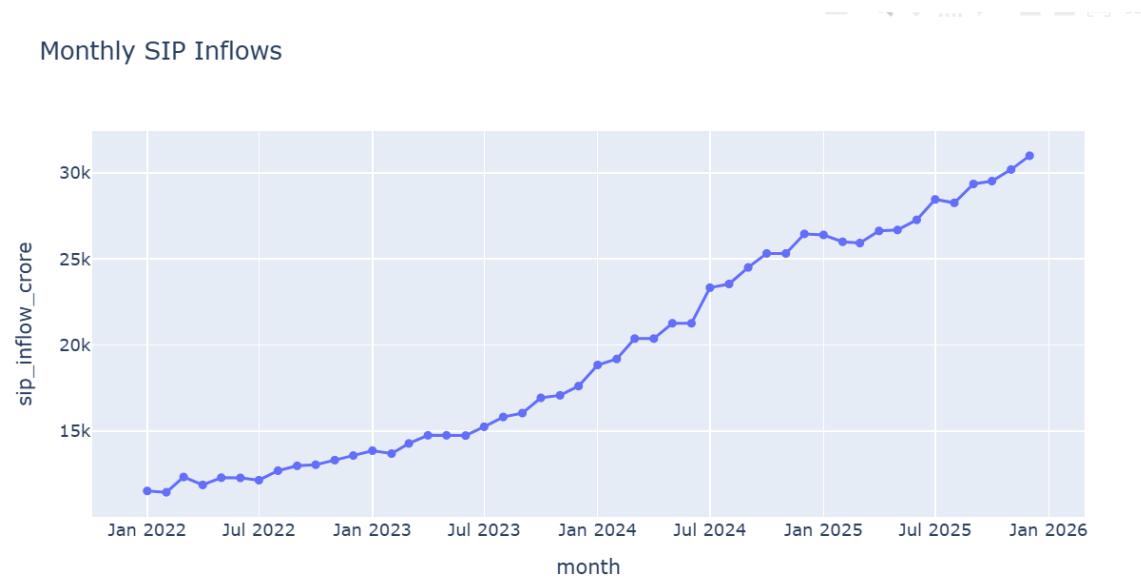
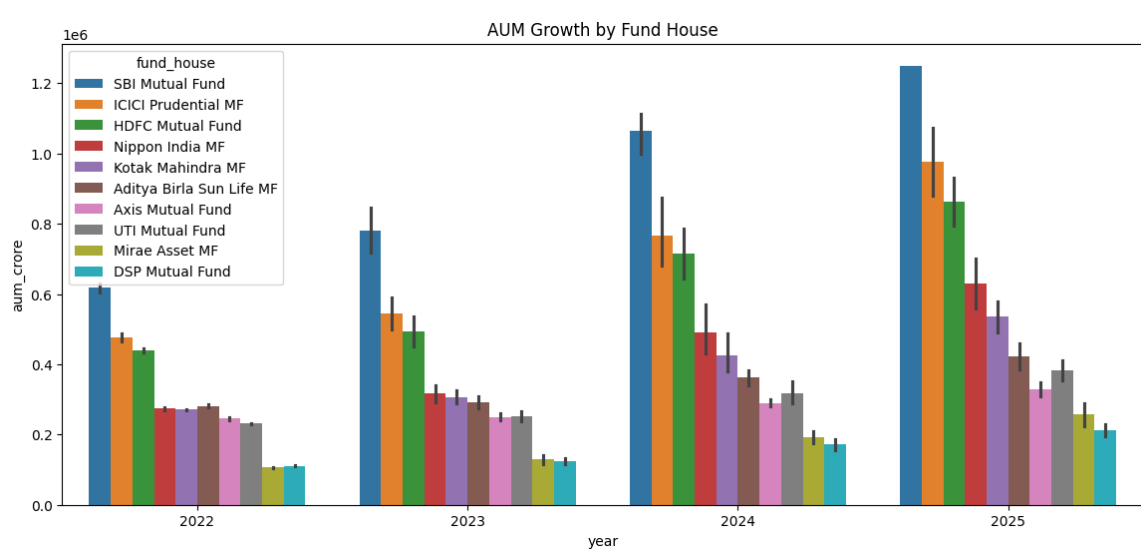
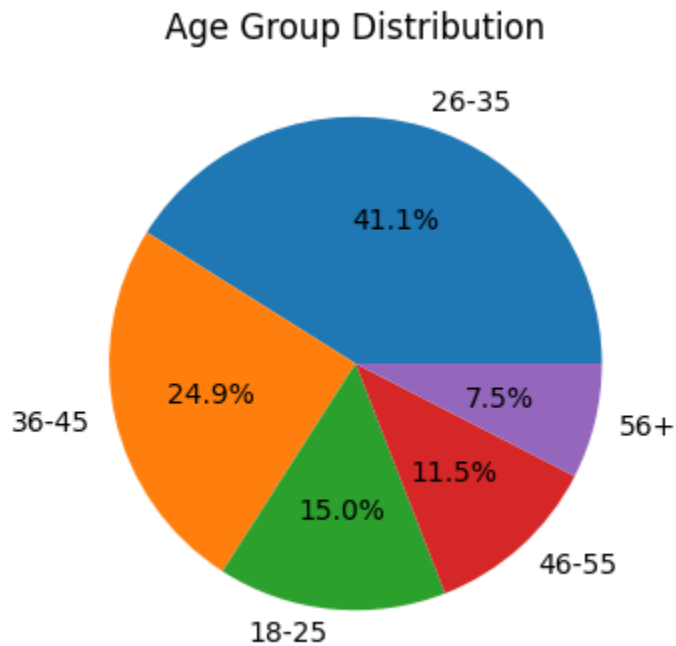


Chart 2: AUM by Fund House



**Chart 3: Age group distribution**



### **Conclusion**

**The EDA findings indicate that mutual fund investments are primarily driven by middle-aged investors, while a few major fund houses dominate the industry in terms of AUM. Additionally, the steady growth in monthly SIP inflows highlights the increasing popularity of systematic investing among retail investors. These insights provide a strong foundation for performance analysis, risk assessment, investor behavior studies, and dashboard reporting.**

# Performance Analytics

Performance analytics was conducted to evaluate the return potential and risk characteristics of mutual fund schemes. Various risk-adjusted and market-relative performance measures were calculated to provide a comprehensive assessment of fund performance.

## Metrics Evaluated

### Sharpe Ratio

Measures the excess return generated per unit of total risk. A higher Sharpe Ratio indicates better risk-adjusted performance.

### Sortino Ratio

Measures return relative to downside risk only. It provides a more focused evaluation of performance by penalizing only negative volatility.

### Alpha

Represents the excess return generated by a fund beyond what would be expected based on its market risk exposure.

### Beta

Measures the sensitivity of a fund's returns to market movements. A Beta greater than 1 indicates higher volatility than the market, while a Beta less than 1 indicates lower volatility.

### CAGR (Compound Annual Growth Rate)

Measures the annualized growth rate of an investment over a specified period, assuming profits are reinvested.

### Volatility

Measures the variability of returns and serves as an indicator of investment risk. Higher volatility implies greater uncertainty in returns.

### Maximum Drawdown

Measures the largest decline from a portfolio's peak value to its lowest point during the analysis period, indicating downside risk exposure.

## Performance Analysis Output



### Key Findings

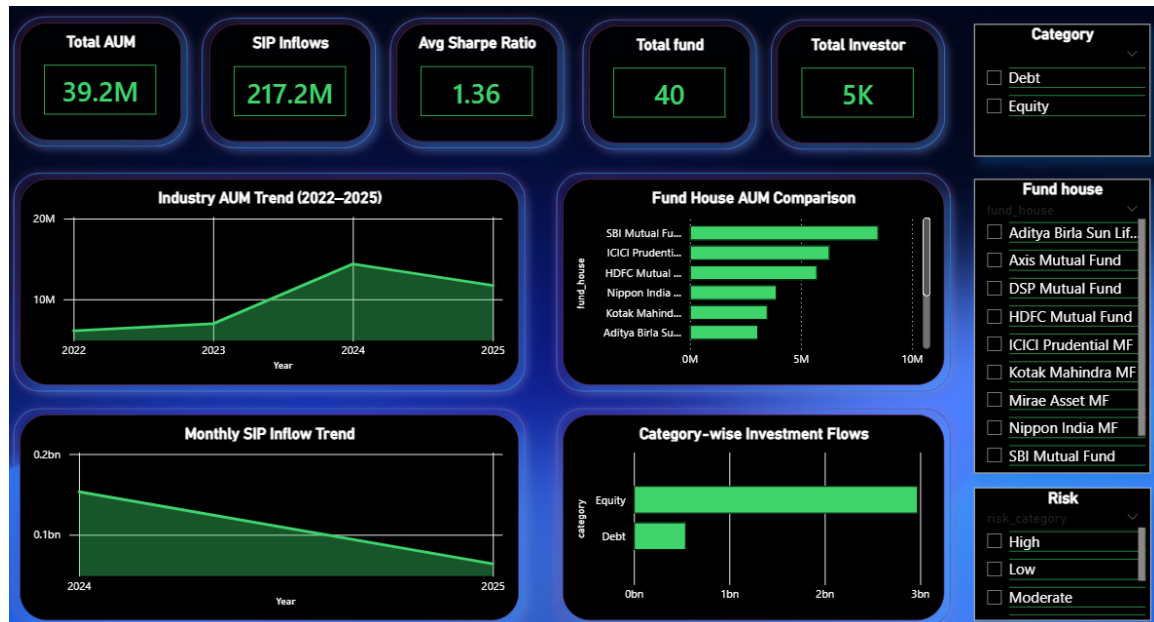
1. Risk-adjusted performance varied across mutual fund schemes, with certain funds delivering superior Sharpe and Sortino Ratios.
2. Alpha analysis identified funds that generated returns above benchmark expectations.
3. Beta values revealed differences in market sensitivity across schemes, highlighting varying risk profiles.
4. Volatility and Maximum Drawdown metrics provided insights into the downside risk associated with each fund.
5. CAGR analysis demonstrated long-term growth potential and enabled comparison of investment performance across schemes.

### Conclusion

The performance analytics framework enabled a comprehensive evaluation of mutual fund schemes by combining return, risk, and market-relative measures. These metrics formed the basis for fund ranking, dashboard visualizations, advanced risk analytics, and the fund recommendation system.

# Power BI Dashboard

## Dashboard Page 1 – Market Overview



### Purpose

This page provides a high-level overview of the mutual fund industry and overall market performance.

### Key Components

- Total Assets Under Management (AUM)
- Total Investor Count
- Total SIP Inflows
- Total Fund Count
- Industry AUM Trend
- Fund House Comparison
- Interactive Slicers

### Key Insights

- The dashboard provides a consolidated view of the mutual fund ecosystem.
- AUM trends help identify overall industry growth.
- Fund house comparison highlights market leaders and competitive positioning.

## Dashboard Page 2 – Fund Performance & Risk

### Purpose

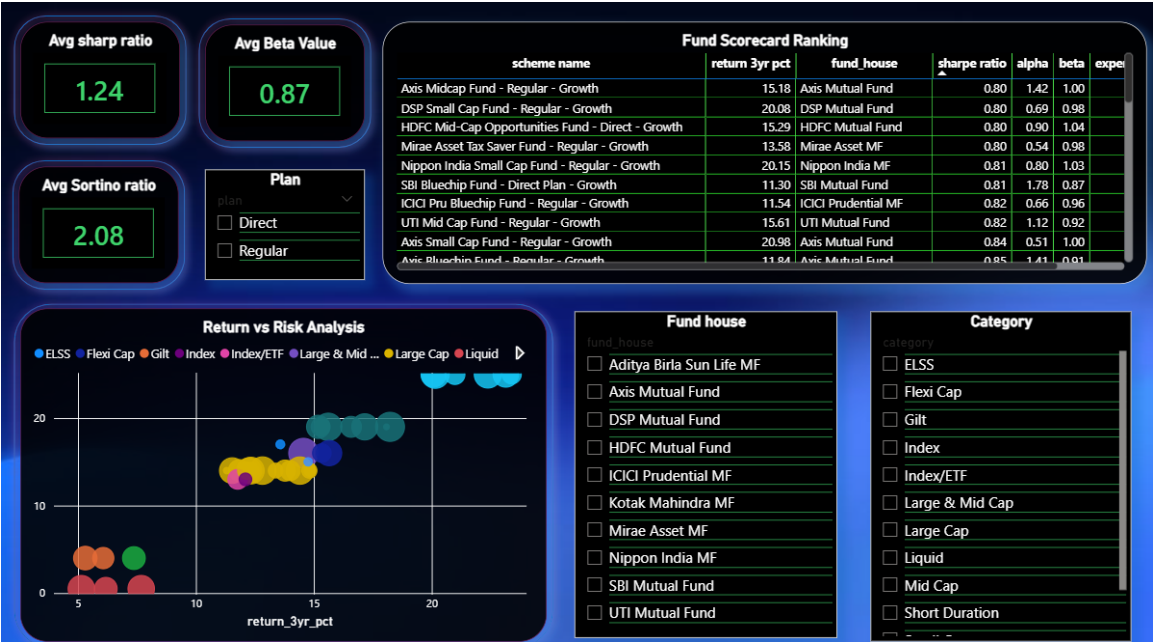
This page evaluates mutual fund schemes using risk-adjusted performance metrics.

### Key Components

- Sharpe Ratio Analysis
- Sortino Ratio Analysis
- Alpha Ranking
- Maximum Drawdown Analysis
- Return vs Risk Comparison
- Fund Scorecard

### Key Insights

- High-performing funds can be identified through superior Sharpe and Sortino Ratios.
- Alpha analysis helps detect funds outperforming benchmark expectations.
- Maximum Drawdown highlights downside risk exposure.
- Return-risk visualization supports informed investment decisions.



## Dashboard Page 3 – Investor Analytics



### Purpose

This page focuses on investor demographics, investment behavior, and transaction patterns.

### Key Components

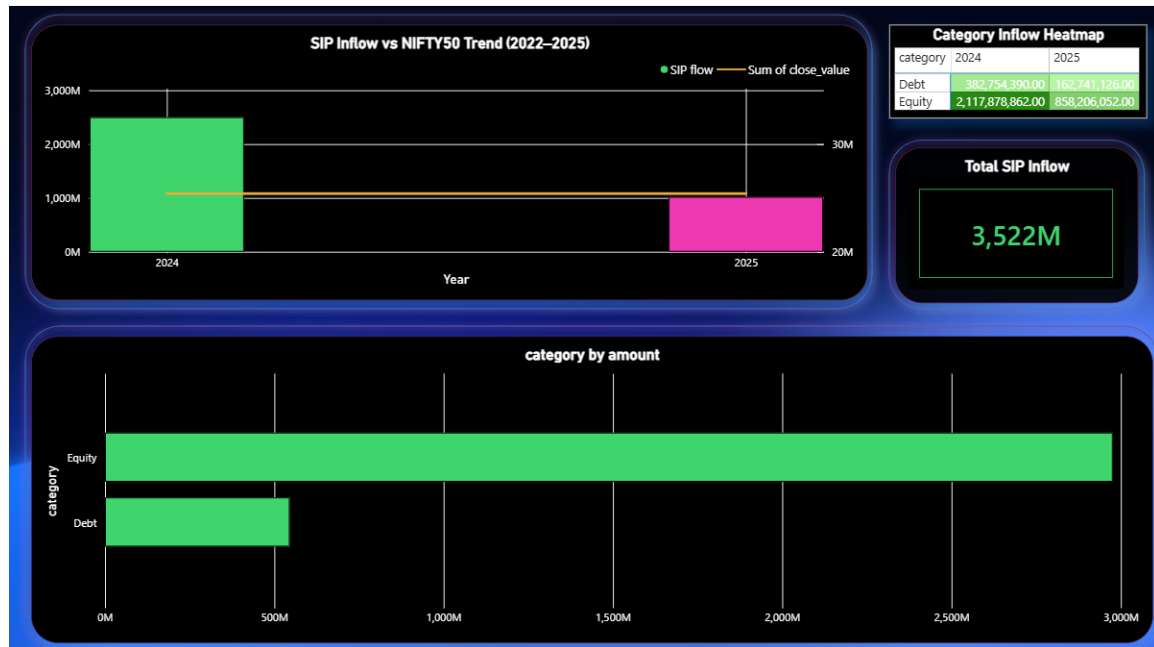
- Age Group Distribution
- State-wise Investment Analysis
- Investor Demographics
- Transaction Trends
- Cohort Analysis Results

### Key Insights

- Middle-aged investors represent the largest investor segment.
- Geographic analysis reveals regions with higher mutual fund participation.
- Transaction trends help understand investor activity patterns.
- Cohort analysis provides insights into investor retention and engagement.



## Dashboard Page 4 – SIP & Market Trends



### Purpose

This page analyzes SIP inflows, benchmark performance, and overall market trends.

### Key Components

- Monthly SIP Inflow Trend
- Benchmark Index Comparison
- Category-wise Investment Flows
- Market Trend Indicators
- Sector Concentration Analysis

### Key Insights

- SIP inflows demonstrate investor confidence and participation trends.
- Benchmark comparison helps evaluate fund performance relative to market indices.
- Category-wise flows reveal investor preferences across fund categories.
- Market trend analysis supports strategic investment decision-making.

# Limitations

## Limitations

1. The analysis is based entirely on historical data; therefore, past performance may not accurately predict future fund performance.
2. Portfolio holdings data was available for only a subset of mutual fund schemes, limiting the scope of sector concentration analysis.
3. Benchmark comparisons were restricted to the available index datasets included in the project.
4. The fund recommendation system uses historical performance metrics and risk classifications, which may not capture future market conditions or investor preferences.
5. Investor behavior analysis was conducted using the available transaction records and may not fully represent all investor activities.
6. The Power BI dashboard relies on periodic data updates and does not provide real-time market information.

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## Summary

Despite certain data and market limitations, the project provides a robust framework for mutual fund performance evaluation, investor analytics, risk assessment, and business intelligence reporting. The assumptions made during analysis are reasonable and consistent with industry-standard analytical practices.

# Recommendations

Based on the analysis conducted throughout the project, the following business and investment recommendations are proposed:

## Business Recommendations

### 1. Strengthen Investor Retention Programs

The SIP Continuity Analysis identified investors with irregular contribution patterns. Fund houses should implement targeted engagement strategies such as reminders, educational campaigns, and personalized communication to improve SIP continuation rates.

### 2. Expand Investor Awareness Initiatives

The demographic analysis indicates lower participation among younger investors. Asset management companies should focus on financial literacy programs and digital marketing campaigns to encourage early investment adoption.

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## Investment Recommendations

### 1. Focus on Risk-Adjusted Returns

Investors should evaluate funds using Sharpe Ratio and Sortino Ratio rather than relying solely on absolute returns, as these metrics provide a better assessment of risk-adjusted performance.

### 2. Diversify Across Funds and Sectors

HHI analysis indicates that some funds are more concentrated than others. Investors should maintain diversified portfolios to reduce concentration risk and improve long-term stability.

### 3. Maintain Long-Term SIP Investments

Consistent SIP investing helps reduce the impact of market volatility and promotes disciplined wealth creation over time.

## Conclusion

The implementation of these recommendations can help improve investor engagement, enhance portfolio management practices, reduce risk exposure, and support better long-term investment outcomes for both investors and asset management companies.

# Conclusion

## Conclusion

The Bluestock Mutual Fund Analytics Capstone Project successfully demonstrated the application of data engineering, financial analytics, database management, and business intelligence techniques to solve real-world investment analysis problems. By integrating multiple datasets including NAV history, investor transactions, portfolio holdings, and benchmark indices, a comprehensive mutual fund analytics platform was developed.

The project involved the complete analytics lifecycle, including data ingestion, cleaning, transformation, database design, exploratory data analysis, performance measurement, advanced risk analytics, and dashboard development. Key financial metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Beta, CAGR, Volatility, and Maximum Drawdown were calculated to evaluate mutual fund performance. Advanced analytics including VaR, CVaR, Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, and Sector HHI Concentration Analysis provided deeper insights into risk, investor behavior, and portfolio diversification.

An interactive four-page Power BI dashboard was developed to enable effective visualization and exploration of market trends, fund performance, investor analytics, and SIP activity. In addition, a risk-based fund recommendation system was implemented to support investment decision-making based on investor risk profiles.

Overall, the project highlights the value of combining Python, SQL, SQLite, and Power BI to transform raw financial data into actionable business insights. The developed solution provides a scalable framework for mutual fund analysis and demonstrates how data-driven approaches can support informed investment and business decisions in the financial services industry.

The successful completion of this project strengthened practical skills in data analytics, financial analysis, database management, visualization, and problem-solving while providing valuable experience in building an end-to-end analytics solution.